Name: _____ **Date:** _____

Software Engineering Specialization

Exercise: Movie Shop—Use-Case Model

- The system must be able to handle both physical and digital videos.

actors & functionality:

- It must be able to record which videos are sold and rented and by whom.

actors & functionality:

- For sold videos, the quantity sold should be recorded; for physical video rental, which copy is rented and when it is due back should be recorded.

actors & functionality:

- The system should keep track of overdue rentals of physical videos and send email notices to customers who have videos overdue.

actors & functionality:

- There will be a customer membership option for an annual fee, which will entitle a member to discounts (10%) on the sale and rental of videos.

actors & functionality:

- Members should be able to make reservations for physical video rentals either in person at the shop, by telephone or via the Web.

actors & functionality:

- A member can reserve at most five physical videos at any one time, but there is no limit on how many physical videos a member or nonmember can rent at any one time.

actors & functionality:

- As an added feature, the shop would like to allow customers (either members or nonmembers) to input, via the Web, mini-reviews (up to 100 words) and a rating (from 1, lowest, to 10, highest) of videos they have purchased or rented.

actors & functionality:

- These reviews should be anonymous if the customer so wishes (i.e., customers can specify whether they want their name to be made known when other customers browse the reviews).

actors & functionality:

- A sales clerk should be able to enter and update the following information about all customers (members or nonmembers): name, address, phone number, age, sex, and email address.

actors & functionality:

- Members are assigned a membership number by the shop when they become members and a password, which allows them to change their personal information and to buy and rent digital videos via the Web.

actors & functionality:

- The shop manager should be able to generate various reports on the sale and rental of videos.

actors & functionality:

- A sales clerk should be able to sell and rent physical videos and process the return of rented physical videos.

actors & functionality:

- When selling or renting physical videos, a sales clerk must be able to look up customer information and determine whether the customer is a member.

actors & functionality:

- A sales clerk must be able to enter basic information about a video (i.e., video id, title, leading actor(s), director, producer, genre, synopsis, release year, running time, selling price, and rental price).

actors & functionality:

Name: _____ **Date:** _____

Software Engineering Specialization

Exercise: Movie Shop—Use-Case Model

1. On the accompanying sheet containing the problem statement, identify all actors and their required functionality. *(Only actors and functionality that are explicitly given in or implied by the requirements statement should be included.)*
2. In the space below group the functionality into use cases and show the uses cases and their related actors in a use-case context diagram on the accompanying sheet.

Actor	Required Functionality

Name: _____ **Date:** _____

Software Engineering Specialization

Exercise: Movie Shop—Use-Case Model Context Diagram

In the space below, show the uses cases and their related actors in a use-case context diagram.